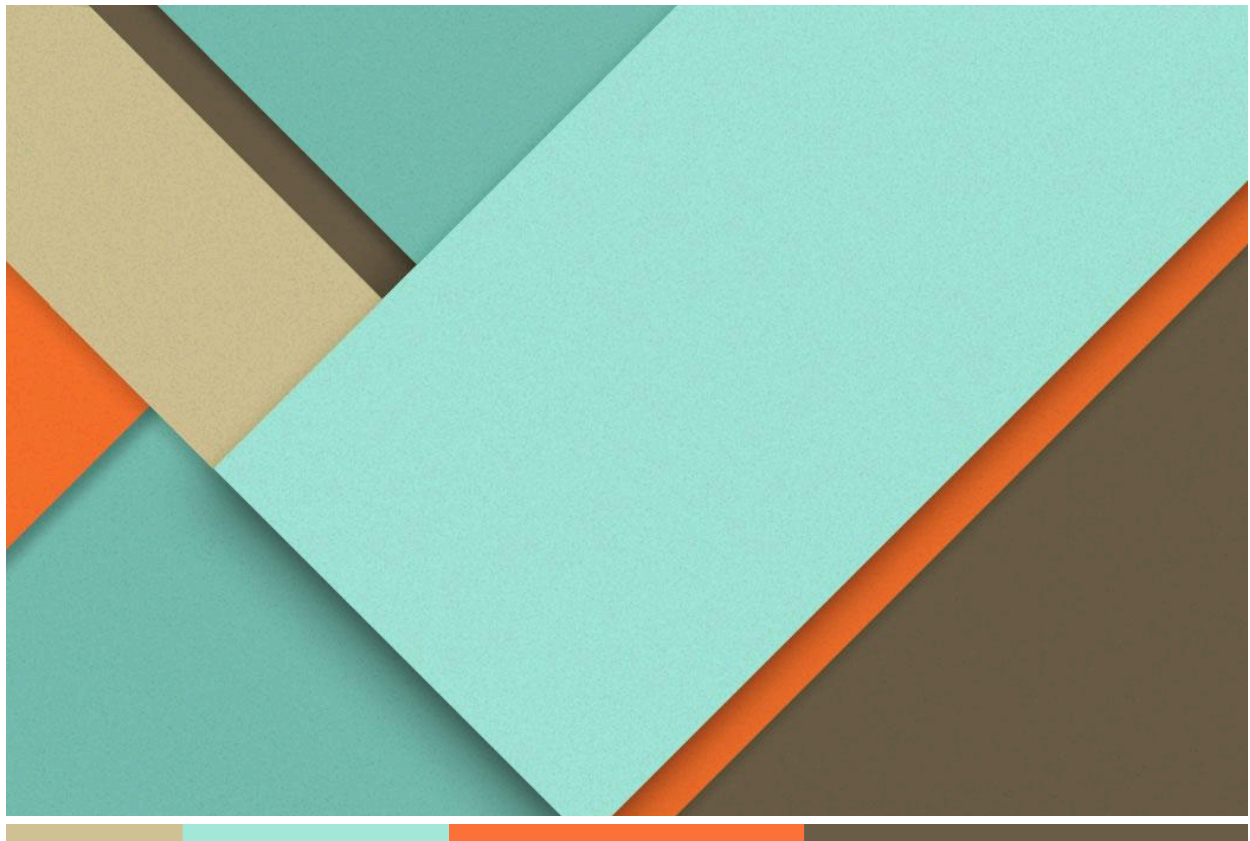




FoodLink

– Connecting Surplus to Need

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Team 4

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Overview

The **Food Rescue and Redistribution System** is a Java-based, multi-role communication platform aimed at minimizing food waste by facilitating structured and trackable collaboration between surplus food donors (e.g., restaurants, supermarkets), food inspectors, logistics volunteers, and shelters. It enables cross-organizational and cross-enterprise coordination through work requests and role-specific functionalities. The platform includes multiple enterprises, each with dedicated organizations and responsibilities, simulating a real-world multi-stakeholder food rescue ecosystem.

Enterprises and Organizations

Enterprises:

1. **FoodChain Inc.** – Handles corporate-level food donations.
2. **VolunteerConnect** – Manages volunteer teams and delivery dispatching.
3. **RescueNet** – Operates shelters and donation centers.
4. **FreshCheck Authority** – Conducts food quality inspections and approval.

Organizations (under each enterprise):

1. WholeFoods Central Org (under FoodChain Inc.)
2. Driver Org (under VolunteerConnect)
3. Task Arrange Org (under VolunteerConnect)
4. Request Collector ORG (under RescueNet)
5. Request Entertain ORG (under RescueNet)
6. NewFood Check ORG (under FreshCheck Authority)

7. PushedFoodCheck ORG (under FreshCheck Authority)

Goals

1. Establish a comprehensive system supporting **8+ distinct, non-admin user roles**.
2. Create role-specific interfaces using Java Swing for secure and intuitive workflows.
3. Facilitate donation request submission, validation, dispatching, delivery, and final confirmation.
4. Track and visualize environmental impact (e.g., food saved, CO₂ reduced).
5. Simulate **cross-organization and cross-enterprise coordination** through work requests.

User Roles (Non-Admin, Unique, ≥8)

1. **Food Provider** – Submits food donation requests.
2. **Volunteer** – Accepts and completes food delivery tasks.
3. **Coordination Supervisor** – Reviews validated donations and assigns delivery tasks.
4. **Shelter Receiver** – Confirms food delivery at shelter level.
5. **Volunteer Scheduler** – Manages volunteer grouping and availability.
6. **Food Quality Inspector** – Evaluates donated food for safety and quality.
7. **Report Analyst** – Views system-wide metrics and environmental reports.
8. **Registration Reviewer** – Reviews new user and organization access requests.

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9. **Shelter Manager** (*optional*) – Oversees shelter resources and internal processes.

Specifications

Implement **role-based access control** through centralized authentication.

The system includes these key entities:

- EcoSystem, Network, Enterprise, Organization, UserAccount, Role, FoodItem, DonationRequest, DeliveryTask, WorkRequest, Report.

Functional Features:

- **Donation submission:** Food Providers input donation details.
- **Inspection workflow:** Food Inspectors evaluate and approve items.
- **Coordination:** Supervisors assign delivery tasks to eligible volunteers.
- **Delivery tracking:** Volunteers manage and update their assigned deliveries.
- **Receiving and confirmation:** Shelter staff confirm the delivery and food quality.
- **Admin-like decisions:** Registration approvers confirm new entity access.
- **Analysis & reporting:** Report Analysts monitor food saved and CO₂ reduced.

Fake data (users, food, tasks) will be auto-generated using the **Java Faker** library.

Work Request Scenarios

At least **4 work requests** will be implemented, including:

1. Cross-Enterprise Request

- A donation from *FoodChain Inc.* is validated and assigned to a volunteer.

2. Cross-Organization Request

- A delivery request is sent from *Volunteer* to *RescueNet*.

3. Quality Evaluation Request

- A donation item is submitted to *FreshCheck Authority* for food inspection before approval.

4. Food Quality Recheck Request

- A Fresh Checker from the FreshCheck Enterprise identifies food items that are borderline or uncertain in quality.

Milestones

Milestone 1: Project Setup and Role Design

- Initialize GitHub + NetBeans project.
- Create: *EcoSystem*, *Network*, *Enterprise*, *Organization*, *UserAccount*, *Role*.
- Implement login and registration flow with role-based routing.

Milestone 2: Food Donation & Validation Module

- Implement *FoodItem*, *DonationRequest*.
- Enable Food Providers to submit donation requests.
- Add validation logic (quantity > 0, valid expiry).

- Route requests to Food Quality Inspectors for approval.

Milestone 3: Coordination & Volunteer Assignment Module

- Coordination Supervisors review approved requests.
- Assign delivery tasks to available volunteers.
- Volunteer Schedulers manage team rosters and availability.

Milestone 4: Delivery & Volunteer Dashboard

- Volunteers access dashboard to accept/track tasks.
- Status flow: Assigned → In Transit → Completed.
- Volunteer task history maintained.

Milestone 5: Receiving & Shelter Operations Module

- Shelter Receivers confirm food arrival & condition.
- Shelter Managers view internal reports and staff overview.
- Food Item and Request status finalized.

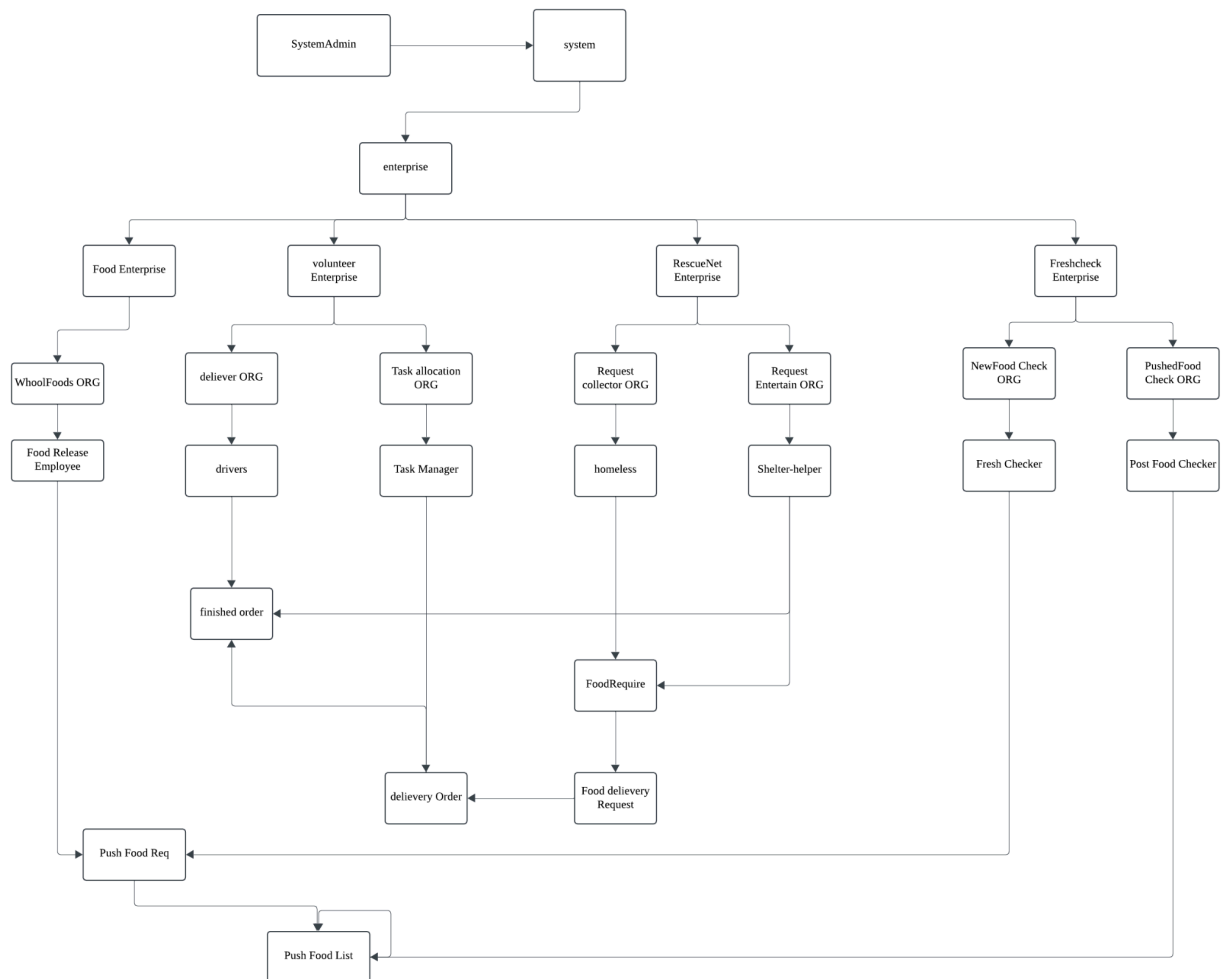
Milestone 6: Reporting & Administrative Tools

- Report Analysts generate daily summary reports (via tables/charts).
- Metrics: total food saved (kg), CO₂ saved, deliveries made.
- Registration Reviewers review incoming user/org access requests.

Milestone 7: Final Integration, Testing, and Presentation

- Integrate all modules into unified Swing UI.
- Populate with **Faker data** (users, food, organizations, roles).
- Test all role workflows (unit + integration).
- Deliverables:
 - Live Demo (20 min)
 - PPT Presentation (5 min)
 - UML & Component Diagrams
 - GitHub submission
 - Code documentation

UML



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